

Statistical Foundations of Machine Learning

Practice Set 1 – Practice Problems

Note: Problems in this section are for practice only. Solutions are not to be submitted.

1. Knockout Tournament

Question. A knockout tournament has k rounds. In each round, half the players are eliminated. Define a suitable sample space for the tournament outcome.

Answer.

Initially there are 2^k players.

A natural sample space is:

$$\Omega = \{\text{all possible complete tournament brackets}\}$$

Each $\omega \in \Omega$ specifies the winner of every match in every round.

Number of matches:

$$2^k - 1$$

Thus,

$$|\Omega| = 2^{2^k - 1}$$

Each element need not be equally likely since match probabilities may differ.

2. Instructors Teaching

Question. Four instructors randomly teach each of N lectures. Find probability all teach at least once. Generalise to k instructors.

Answer.

Total assignments:

$$4^N$$

Using inclusion–exclusion:

$$P(\text{all teach}) = 1 - \binom{4}{1} \left(\frac{3}{4}\right)^N + \binom{4}{2} \left(\frac{2}{4}\right)^N - \binom{4}{3} \left(\frac{1}{4}\right)^N$$

General case (k instructors):

$$P = \sum_{j=0}^k (-1)^j \binom{k}{j} \left(\frac{k-j}{k}\right)^N$$

3. Numbers on a Circle

Question. Numbers 1, 2, 3, 4, 5, and 6 are randomly placed on a circle. What is the probability that they are placed in increasing order?

Answer.

All circular arrangements of 6 distinct numbers are equally likely.

The number of distinct circular permutations of 6 elements is:

$$(6 - 1)! = 5! = 120$$

Now consider arrangements that are increasing clockwise. If the numbers are in increasing order around the circle, the smallest number (1) can be in any of the 6 positions, and the remaining numbers must follow in order.

Thus there are 6 favorable arrangements.

Therefore,

$$P = \frac{6}{120} = \frac{1}{20}.$$

4. Dice Game

Question. A and B roll dice with B re-rolling once if first roll is 1. A wins ties.

Answer.

Distribution of B:

If B rolls 1 initially, he re-rolls.

$$P(B = 1) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

For $k = 2, \dots, 6$:

$$P(B = k) = P(\text{first} = k) + P(\text{first} = 1, \text{second} = k)$$

$$= \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{6} + \frac{1}{36} = \frac{7}{36}$$

Now compute:

$$P(A \geq B) = \sum_{b=1}^6 P(B = b)P(A \geq b)$$

We use:

$$P(A \geq 1) = 1, P(A \geq 2) = \frac{5}{6}, P(A \geq 3) = \frac{4}{6}, P(A \geq 4) = \frac{3}{6}, P(A \geq 5) = \frac{2}{6}, P(A \geq 6) = \frac{1}{6}$$

Thus,

$$\begin{aligned} P(A \geq B) &= \frac{1}{36}(1) + \frac{7}{36} \left(\frac{5}{6} + \frac{4}{6} + \frac{3}{6} + \frac{2}{6} + \frac{1}{6} \right) \\ &= \frac{1}{36} + \frac{7}{36} \cdot \frac{15}{6} = \frac{1}{36} + \frac{35}{72} = \frac{37}{72} \end{aligned}$$

$$\boxed{P(A \text{ wins}) = \frac{37}{72}}$$

5. Urns

Question. n urns; r -th urn has $(r - 1)$ brown and $(n - r)$ black balls. Pick urn uniformly; draw two balls without replacement.

Answer.

For urn r :

$$P(\text{second black}) = \frac{n - r}{n - 1}$$

Averaging over r :

$$P(\text{second black}) = \frac{1}{n} \sum_{r=1}^n \frac{n - r}{n - 1} = \frac{1}{2}$$

Also,

$$P(\text{second black} \mid \text{first black}) = \frac{1}{2}$$

6. Inclusion–Exclusion

Question. Uniformly choose number in $[1, 10^6]$. Probability divisible by 4, 6, or 9?

Answer.

Let $N = 10^6$. Using inclusion–exclusion:

$$P = \frac{\lfloor \frac{N}{4} \rfloor + \lfloor \frac{N}{6} \rfloor + \lfloor \frac{N}{9} \rfloor - \lfloor \frac{N}{12} \rfloor - \lfloor \frac{N}{36} \rfloor - \lfloor \frac{N}{18} \rfloor + \lfloor \frac{N}{36} \rfloor}{N}$$

So the $\lfloor N/36 \rfloor$ terms cancel, giving:

$$P = \frac{\lfloor \frac{N}{4} \rfloor + \lfloor \frac{N}{6} \rfloor + \lfloor \frac{N}{9} \rfloor - \lfloor \frac{N}{12} \rfloor - \lfloor \frac{N}{18} \rfloor}{N}$$

Compute:

$$\left\lfloor \frac{10^6}{4} \right\rfloor = 250000, \quad \left\lfloor \frac{10^6}{6} \right\rfloor = 166666, \quad \left\lfloor \frac{10^6}{9} \right\rfloor = 111111$$
$$\left\lfloor \frac{10^6}{12} \right\rfloor = 83333, \quad \left\lfloor \frac{10^6}{18} \right\rfloor = 55555$$

Thus total count = 388889, hence

$$P = \frac{388889}{10^6} = 0.388889.$$

7. Polya Urn

Question. Start with 1 black, 1 brown. After N reinforcement steps, find probability of k black balls.

Answer.

This is a Polya urn model.

$$P(k \text{ black}) = \frac{1}{N+1} \quad k = 1, 2, \dots, N+1$$

Uniform distribution.

8. Best of Five

Question. Evenly matched players. Probability loser wins k sets?

Answer.

For $k = 0, 1, 2$:

$$P = \binom{2+k}{k} \left(\frac{1}{2}\right)^{3+k}$$

9. Coin Rewards

Question. A_k = reward at time k if T followed by H .

(a) A_k and A_{k+1} ?

Event A_k depends on tosses $(k, k+1)$, while A_{k+1} depends on tosses $(k+1, k+2)$.

They share the common toss at time $k+1$, so the events are not independent.

(b) A_k and A_{k+2} ?

Event A_k depends on tosses $(k, k+1)$, while A_{k+2} depends on tosses $(k+2, k+3)$.

There is no overlap in tosses. Since coin tosses are independent, the events are independent.

10. Coin Types (Bayes)

Total coins = 5.

$$P(A) = \frac{2}{5}, \quad P(B) = \frac{2}{5}, \quad P(C) = \frac{1}{5}$$

$$P(H) = \frac{2}{5}(0.5) + \frac{2}{5}(0.7) + \frac{1}{5}(0.9) = 0.66$$

Bayes:

$$P(A|H) = \frac{0.2}{0.66}$$

$$P(B|H) = \frac{0.28}{0.66}$$

$$P(C|H) = \frac{0.18}{0.66}$$

11. Inequalities

(a)

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

(b)

Apply induction:

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1)$$

12. High Probability Events

Given:

$$P(A) > 1 - \delta, \quad P(B) > 1 - \delta$$

$$P(A \cap B) \geq P(A) + P(B) - 1 > 1 - 2\delta$$

Thus close to 1 for small δ .

13. Erdős–Rényi $G(n, p)$

For a fixed node:

$$P(\text{isolated}) = (1 - p)^{n-1}$$

By union bound:

$$P(B_n) \leq n(1 - p)^{n-1}$$

If

$$p = \frac{(1 + \epsilon) \ln n}{n}$$

Then

$$n(1 - p)^{n-1} \sim ne^{-(1+\epsilon)\ln n} = n^{-\epsilon} \rightarrow 0$$

Hence

$$\lim_{n \rightarrow \infty} P(B_n) = 0$$

14. Coupon Collector (General Probabilities)

Question. Each coupon independently has type i with probability p_i , $\sum p_i = 1$. After k coupons, let A_i = at least one coupon of type i .

Answer.

(a)

$$P(A_i) = 1 - (1 - p_i)^k$$

(b)

$$P(A_i \cup A_j) = 1 - (1 - p_i)^k - (1 - p_j)^k + (1 - p_i - p_j)^k$$

(c)

$$P(A_i|A_j) = \frac{P(A_i \cap A_j)}{P(A_j)}$$

$$P(A_i \cap A_j) = 1 - (1 - p_i)^k - (1 - p_j)^k + (1 - p_i - p_j)^k$$

15. Monty Hall Problem

Question. Show $P(E_1) < P(E_2|C)$.

Answer.

Initial:

$$P(E_1) = \frac{1}{3}$$

If initial guess wrong ($2/3$), switching wins.

Thus:

$$P(E_2|C) = \frac{2}{3}$$

Hence

$$\frac{1}{3} < \frac{2}{3}$$

16. Hat Problem (Derangements)

Question. Probability no one gets own hat.

Answer.

$$P_n = \frac{!n}{n!} = \sum_{k=0}^n \frac{(-1)^k}{k!}$$

As $n \rightarrow \infty$:

$$P_n \rightarrow e^{-1}$$

17. Bounds on p and q

Given:

$$P\left(\bigcup_{n=1}^N A_n\right) = 1$$

Assume symmetry:

$$P(A_i) = p \quad \text{for all } i,$$

$$P(A_i \cap A_j) = q \quad \text{for all } i \neq j,$$

and assume all triple (and higher) intersections are zero.

Using inclusion-exclusion up to pairwise terms,

$$P\left(\bigcup_{n=1}^N A_n\right) = \sum_{i=1}^N P(A_i) - \sum_{i < j} P(A_i \cap A_j)$$

$$1 = Np - \binom{N}{2}q$$

Since probabilities are nonnegative,

$$Np \geq 1 \quad \Rightarrow \quad p \geq \frac{1}{N}.$$

Also,

$$\binom{N}{2}q \leq Np \quad \Rightarrow \quad q \leq \frac{2}{N}.$$

18. Birthday Problem (3 People)

$$P(\text{all distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365}$$

$$P(\text{at least two equal}) = 1 - \frac{364 \cdot 363}{365^2} \approx 0.0082$$

19. Hat-Themed Party

$$\begin{aligned} P\left(\bigcup_{i=1}^N A_i\right) &= 1 - P(\text{no fixed points}) \\ &= 1 - \frac{!N}{N!} \end{aligned}$$

As $N \rightarrow \infty$:

$$\rightarrow 1 - \frac{1}{e}$$

20. Extremal Values

Given:

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}$$

$$P(A \cap B)_{\max} = \frac{1}{4}$$

$$P(A \cap B)_{\min} = 0$$

$$P(A \cup B)_{\max} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$P(A \cup B)_{\min} = \frac{1}{2}$$

21. Bayes (Weather)

$$P(H) = 0.4, \quad P(C) = 0.6$$

$$P(R|H) = 0.2, \quad P(R|C) = 0.7$$

$$P(R) = 0.4(0.2) + 0.6(0.7) = 0.5$$

$$P(H|R) = \frac{0.4 \cdot 0.2}{0.5} = 0.16$$

22. Law of Total Probability

Given:

$$P(A_i) = \frac{1}{2^i}, \quad P(C|A_i) = \frac{1}{3^i}$$

$$\begin{aligned} P(C) &= \sum_{i=1}^{\infty} P(C|A_i)P(A_i) = \sum \frac{1}{6^i} \\ &= \frac{1/6}{1 - 1/6} = \frac{1}{5} \end{aligned}$$

23. Conditional Probability Inequality

Given:

$$P(A|B) > P(A)$$

$$\frac{P(A \cap B)}{P(B)} > P(A) \Rightarrow P(A \cap B) > P(A)P(B)$$

Thus:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} > P(B)$$

For the complement, note that if B makes A more likely than usual, then the absence of B must make A less likely than usual. Formally,

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

Since $P(A|B) > P(A)$, the weighted average can equal $P(A)$ only if

$$P(A|B^c) < P(A).$$

24. Random Integer + Coin

$$P(I = n) = \frac{1}{2^n}$$

$$\begin{aligned} P(H) &= \sum_{n=1}^{\infty} \frac{1}{2^n} e^{-n} = \sum \left(\frac{e^{-1}}{2} \right)^n \\ &= \frac{e^{-1}/2}{1 - e^{-1}/2} \end{aligned}$$

25. Disjointness

Given:

$$\sum_{i=1}^n P(A_i) = P\left(\bigcup_{i=1}^n A_i\right)$$

In general,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

(with equality only if there is no overlap).

Thus equality holds only when all pairwise intersections are zero:

$$P(A_i \cap A_j) = 0 \quad \text{for all } i \neq j.$$

Hence the events are pairwise disjoint.

26. Truthful Man (Bayes)

$$P(T) = \frac{3}{4}, \quad P(L) = \frac{1}{4}$$

If the man lies, he chooses uniformly among the other five numbers.

$$P(\text{report } 6) = P(T)P(6|T) + P(L)P(6|L)$$

$$= \frac{3}{4} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{1}{5} = \frac{1}{8} + \frac{1}{20} = \frac{7}{40}$$

$$P(\text{actual } 6 | \text{report } 6) = \frac{(3/4)(1/6)}{7/40} = \frac{1/8}{7/40} = \frac{5}{7}$$